

Practical Demo: Hedging a Bond Portfolio with Treasury Futures

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Learning Objectives

After completing this demo you should be able to:

- calculate a bond portfolio's analytical and Treasury-beta-adjusted DV01;
- derive Treasury futures DV01 from the cheapest-to-deliver (CTD) bond and conversion factor;
- size and round a short futures hedge;
- compare portfolio P&L before and after the hedge; and
- identify residual basis, spread, curve, rounding, and CTD risks.

1 Market Inputs Versus Illustrative Inputs

The workflow resembles a daily risk-management task, but every number below is illustrative rather than live market data. A live hedge would use current positions, clean prices, accrued interest, key-rate durations, deliverable-basket data, conversion factors, CTD candidates, financing rates, and exchange contract specifications.

Simplifying assumptions:

- changes are small enough for first-order DV01 approximations;
- the selected CTD remains unchanged;
- futures DV01 is constant over the scenario;
- the Treasury move is parallel;
- spread duration equals modified duration for the corporate positions;
- yield betas summarize how each position historically responds to Treasury yields; and
- transaction costs, margin cash flows, convexity, carry, and roll are ignored.

Workflow stage	In this case
Observed or supplied	Positions, CTD price, conversion factor, and futures price
Estimated	Treasury yield betas and CTD-based futures DV01
Solved	Integer number of contracts to short
Diagnosed	Residual rate DV01, spread risk, and rounding error
Used for	Before-and-after hedge P&L under a joint rate/spread shock

2 Build the Portfolio Risk Table

The portfolio contains Treasuries and investment-grade corporate bonds. Analytical DV01 is

$$DV01_i = D_{\text{mod},i} \times MV_i \times 0.0001.$$

For cross-hedging, a Treasury yield beta maps each bond's historical price sensitivity to the hedging instrument:

$$DV01_i^{\text{Treasury}} = DV01_i \times \beta_i.$$

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

portfolio = pd.DataFrame({
    "Position": [
        "5-year Treasury notes",
        "10-year Treasury notes",
        "A industrial bonds",
        "BBB financial bonds",
    ],
    "Market value ($mm)": [30.0, 25.0, 24.0, 18.0],
    "Modified duration": [4.5, 7.6, 6.1, 7.2],
    "Treasury yield beta": [1.00, 1.00, 0.88, 0.72],
    "Spread shock in scenario (bp)": [0.0, 0.0, 12.0, 20.0],
})

portfolio["Analytical DV01 ($/bp)"] = (
    portfolio["Market value ($mm)"] * 1_000_000
    * portfolio["Modified duration"] * 0.0001
)
portfolio["Treasury DV01 ($/bp)"] = (
    portfolio["Analytical DV01 ($/bp)"] * portfolio["Treasury yield beta"]
)
portfolio["Spread DV01 ($/bp)"] = np.where(
    portfolio["Position"].str.contains("bonds"),
    portfolio["Analytical DV01 ($/bp)"],
    0.0,
)

portfolio.round(2)

```

	Position	Market value (\$mm)	Modified duration	Treasury yield beta	Spread shock in
0	5-year Treasury notes	30.0	4.5	1.00	0.0
1	10-year Treasury notes	25.0	7.6	1.00	0.0
2	A industrial bonds	24.0	6.1	0.88	12.0
3	BBB financial bonds	18.0	7.2	0.72	20.0

```

analytical_dv01 = portfolio["Analytical DV01 ($/bp)"].sum()
treasury_dv01 = portfolio["Treasury DV01 ($/bp)"].sum()

```

```
spread_dv01 = portfolio["Spread DV01 ($/bp)"].sum()

pd.Series({
    "Portfolio market value ($mm)": portfolio["Market value ($mm)"].sum(),
    "Analytical DV01 ($/bp)": analytical_dv01,
    "Treasury-beta-adjusted DV01 ($/bp)": treasury_dv01,
    "Corporate spread DV01 ($/bp)": spread_dv01,
}).round(2)
```

```
Portfolio market value ($mm)          97.0
Analytical DV01 ($/bp)              60100.0
Treasury-beta-adjusted DV01 ($/bp)   54714.4
Corporate spread DV01 ($/bp)        27600.0
dtype: float64
```

The analytical DV01 describes a common change in every position's discount yield. The beta-adjusted DV01 is the exposure targeted by a Treasury futures hedge. Spread risk remains a separate exposure.

3 Frame the CTD and Conversion-Factor Problem

Treasury futures are linked to a delivery basket. The short chooses the eligible security to deliver, and the cheapest-to-deliver issue often determines the futures contract's effective interest-rate sensitivity.

Suppose the current CTD candidate has the following illustrative inputs:

Input	Value
CTD clean price	101.75
CTD modified duration	7.35
Deliverable face amount	\$100,000
Conversion factor	0.9275
Futures price	112-16 = 112.50

Ignoring accrued interest and delivery-option details,

$$DV01_{CTD} = D_{\text{mod},CTD} P_{CTD}(0.0001),$$

and

$$DV01_{futures} \approx \frac{DV01_{CTD}}{CF}.$$

```
ctd_clean_price = 101.75
ctd_modified_duration = 7.35
deliverable_face = 100_000
conversion_factor = 0.9275
futures_price = 112.50

ctd_market_value = ctd_clean_price / 100 * deliverable_face
ctd_dv01 = ctd_modified_duration * ctd_market_value * 0.0001
futures_dv01 = ctd_dv01 / conversion_factor
approx_invoice_price = futures_price / 100 * deliverable_face * conversion_factor

pd.Series({
    "CTD market value ($)": ctd_market_value,
    "Approximate invoice price before accrued interest ($)": approx_invoice_price,
    "CTD DV01 ($/bp)": ctd_dv01,
    "Futures DV01 ($/bp)": futures_dv01,
}).round(2)
```

CTD market value (\$)	101750.00
Approximate invoice price before accrued interest (\$)	104343.75
CTD DV01 (\$/bp)	74.79
Futures DV01 (\$/bp)	80.63
dtype:	float64

The futures contract does not have a true bond duration. Its practical sensitivity is inherited from the current CTD bond and scaled by the conversion factor.

4 Size and Round the Hedge

The portfolio is long bonds and loses when rates rise, so the hedge shorts Treasury futures. The exact contract count is

$$N^* = \frac{DV01_{portfolio}^{Treasury}}{DV01_{futures}}.$$

```

exact_contracts = treasury_dv01 / futures_dv01
rounded_contracts = int(np rint(exact_contracts))
hedge_dv01 = rounded_contracts * futures_dv01
residual_treasury_dv01 = treasury_dv01 - hedge_dv01
hedge_ratio = hedge_dv01 / treasury_dv01

pd.Series({
    "Exact contracts to short": exact_contracts,
    "Rounded contracts to short": rounded_contracts,
    "Hedge DV01 ($/bp)": hedge_dv01,
    "Residual Treasury DV01 ($/bp)": residual_treasury_dv01,
    "Hedge ratio": hedge_ratio,
}).round(4)

```

Exact contracts to short	678.5687
Rounded contracts to short	679.0000
Hedge DV01 (\$/bp)	54749.1792
Residual Treasury DV01 (\$/bp)	-34.7792
Hedge ratio	1.0006
dtype:	float64

Rounding creates a small residual exposure. A negative residual means the rounded hedge is slightly too large; a positive residual means it is slightly too small.

5 Before-and-After Scenario P&L

Suppose Treasury yields rise by 35 basis points. Corporate spreads also widen by the amounts in the portfolio table. The first-order bond P&L is separated into Treasury and spread components:

$$\Delta V_{portfolio} \approx -DV01^{Treasury} \Delta y_{Treasury} - DV01^{spread} \Delta s.$$

Because the hedge is short futures, its approximate gain when Treasury yields rise is

$$\Delta V_{futures} \approx N \times DV01_{futures} \times \Delta y_{Treasury}.$$

```

treasury_shock_bp = 35.0

portfolio["Treasury P&L ($)"] = -portfolio["Treasury DV01 ($/bp)"] * treasury_shock_bp
portfolio["Spread P&L ($)"] = (
    -portfolio["Spread DV01 ($/bp)"]
    * portfolio["Spread shock in scenario (bp)"]
)
portfolio["Total unhedged P&L ($)"] = (
    portfolio["Treasury P&L ($)"] + portfolio["Spread P&L ($)"]
)

futures_pnl = rounded_contracts * futures_dv01 * treasury_shock_bp
unhedged_pnl = portfolio["Total unhedged P&L ($)"].sum()
hedged_pnl = unhedged_pnl + futures_pnl
residual_rate_pnl = -residual_treasury_dv01 * treasury_shock_bp
spread_pnl = portfolio["Spread P&L ($)"].sum()

portfolio[[
    "Position", "Treasury P&L ($)", "Spread P&L ($)", "Total unhedged P&L ($)"
]].round(0)

```

	Position	Treasury P&L (\$)	Spread P&L (\$)	Total unhedged P&L (\$)
0	5-year Treasury notes	-472500.0	-0.0	-472500.0
1	10-year Treasury notes	-665000.0	-0.0	-665000.0
2	A industrial bonds	-450912.0	-175680.0	-626592.0
3	BBB financial bonds	-326592.0	-259200.0	-585792.0

```

pd.Series({
    "Unhedged portfolio P&L ($)": unhedged_pnl,
    "Short futures P&L ($)": futures_pnl,
    "Hedged portfolio P&L ($)": hedged_pnl,
    "Residual rate P&L from rounding ($)": residual_rate_pnl,
    "Unhedged spread P&L ($)": spread_pnl,
}).round(0)

```

```

Unhedged portfolio P&L ($)      -2349884.0
Short futures P&L ($)           1916221.0
Hedged portfolio P&L ($)        -433663.0
Residual rate P&L from rounding ($)  1217.0

```

```
Unhedged spread P&L ($)                -434880.0
dtype: float64
```

The futures hedge removes almost all of the modeled Treasury-rate loss. It does not hedge the corporate spread widening, which remains the dominant loss after the hedge.

```
treasury_scenarios = np.arange(-75, 76, 5)
unhedged_rate_pnl = -treasury_dv01 * treasury_scenarios
hedged_rate_pnl = unhedged_rate_pnl + hedge_dv01 * treasury_scenarios

fig, ax = plt.subplots(figsize=(7.5, 4.1))
ax.plot(treasury_scenarios, unhedged_rate_pnl / 1_000, linewidth=2.1,
        label="Unhedged Treasury-rate P&L")
ax.plot(treasury_scenarios, hedged_rate_pnl / 1_000, linewidth=2.1,
        label="After rounded futures hedge")
ax.axhline(0, color="black", linewidth=0.8)
ax.axvline(0, color="gray", linestyle="--", linewidth=1)
ax.set(xlabel="Parallel Treasury yield shock (bp)", ylabel="Rate-driven P&L ($ thousands)",
        title="First-order Treasury-rate hedge")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()
```

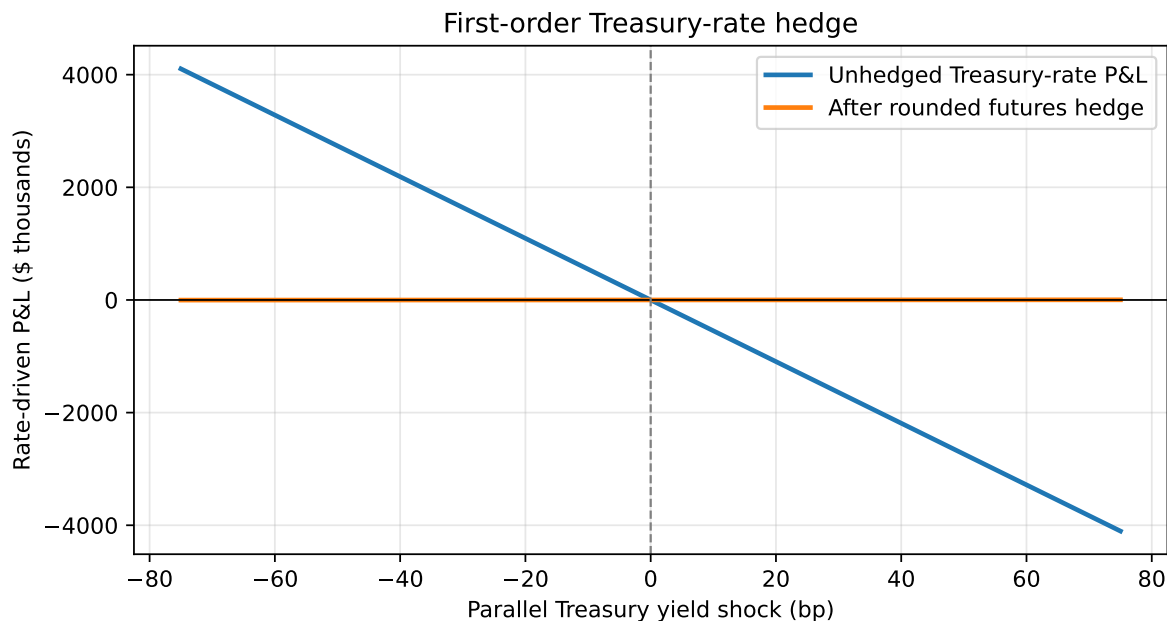



Figure 1: The futures hedge offsets parallel Treasury-rate risk but not spread risk

6 Validation Checks

```
assert ctd_dv01 > 0 and futures_dv01 > 0
assert rounded_contracts > 0
assert abs(treasury_dv01 - hedge_dv01 - residual_treasury_dv01) < 1e-8
assert abs(hedged_pnl - (spread_pnl + residual_rate_pnl)) < 1e-8
assert abs(residual_treasury_dv01) <= 0.5 * futures_dv01 + 1e-10

checks = pd.Series({
    "Futures DV01 is positive": futures_dv01 > 0,
    "Long-bond hedge requires short futures": rounded_contracts > 0,
    "Rounded residual is no more than half one contract DV01": (
        abs(residual_treasury_dv01) <= 0.5 * futures_dv01
    ),
    "Hedged P&L reconciles to spread plus residual-rate P&L": (
        abs(hedged_pnl - spread_pnl - residual_rate_pnl) < 1e-8
    ),
})
checks
```

Futures DV01 is positive	True
Long-bond hedge requires short futures	True
Rounded residual is no more than half one contract DV01	True
Hedged P&L reconciles to spread plus residual-rate P&L	True
dtype: bool	

7 What Can Break the Hedge?

- **Basis risk:** cash bonds and the futures contract may not move together.
- **Curve risk:** one CTD-based DV01 cannot hedge every key-rate exposure.
- **Spread risk:** Treasury futures do not directly hedge corporate credit spreads.
- **Yield-beta error:** historical betas can change in stress periods.
- **CTD switching:** a different deliverable bond can become cheapest to deliver.
- **Conversion-factor and delivery-option effects:** the simple DV01 scaling is approximate.
- **Convexity:** large yield moves make a first-order DV01 hedge less accurate.
- **Rounding, liquidity, margin, and roll:** implementation affects realized results.

8 Financial Interpretation

A Treasury futures hedge is a risk-matching exercise, not a notional-matching exercise. The correct sequence is:

1. identify the portfolio risk to hedge;
2. map that risk into Treasury DV01;
3. estimate futures DV01 from the current CTD and conversion factor;
4. choose and round the contract count; and
5. test the combined position under rate, spread, curve, and basis scenarios.

The hedge is successful here because it neutralizes the stated parallel Treasury-rate exposure. The remaining loss is economically meaningful: it is primarily credit spread risk, which was never within the hedge's scope.